Problem

Let $n, p > 1$ be positive integers, where p is prime. Suppose that

$$n \mid (p-1)$$

and

$$p \mid (n^3 - 1).$$

Prove that

$$4p - 3$$

is a perfect square.

Source: **Iran Mathematical Olympiad 2005**

§1 Extracting Information from the Divisibility Conditions

Since

$$n \mid (p-1),$$

there exists a positive integer k such that

$$p = nk + 1.$$

In particular,

$$p \geq n + 1.$$

Next, from

$$p \mid (n^3 - 1),$$

we factor

$$n^3 - 1 = (n-1)(n^2 + n + 1).$$

Because p is prime, Euclid's lemma implies that

$$p \mid (n-1) \qquad \text{or} \qquad p \mid (n^2 + n + 1).$$

We claim that the first possibility is impossible.

Indeed, if $p \mid (n-1)$, then

$$p \leq n - 1.$$

But from $n \mid (p-1)$ we already have

$$p \geq n + 1,$$

a contradiction.

Therefore,

$$p \mid (n^2 + n + 1).$$

Since $n^2 + n + 1 > 0$, it follows that

$$p \leq n^2 + n + 1.$$

§2 Forcing the Value of p

Recall that

$$p = nk + 1.$$

Since $p \mid (n^2 + n + 1)$, we also have

$$nk + 1 \mid n^2 + n + 1.$$

Multiplying the latter number by k preserves divisibility, so

$$nk + 1 \mid k(n^2 + n + 1).$$

Now compute

$$\begin{aligned} k(n^2 + n + 1) - n(nk + 1) &= kn^2 + kn + k - n^2k - n \\ &= kn + k - n. \end{aligned}$$

Hence

$$nk + 1 \mid (kn + k - n).$$

Since the divisor is positive, we must have

$$nk + 1 \le kn + k - n.$$

Rearranging gives

$$1 \le k - n,$$

so

$$k \ge n + 1.$$

Substituting into $p = nk + 1$ yields

$$p \ge n(n + 1) + 1 = n^2 + n + 1.$$

But earlier we proved that

$$p \le n^2 + n + 1.$$

Therefore,

$$p = n^2 + n + 1.$$

§3 The Final Identity

Now compute:

$$\begin{aligned} 4p - 3 &= 4(n^2 + n + 1) - 3 \\ &= 4n^2 + 4n + 1 \\ &= (2n + 1)^2. \end{aligned}$$

Thus

$$4p - 3$$

is a perfect square.

Indeed,

$$\boxed{4p - 3 = (2n + 1)^2}.$$

This completes the proof.